

10. Are these system specifications consistent? "Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."

Soln: p : "The system software is being upgraded." q : "Users cannot access the file system." r : "User can ^{save} new files."

① $p \rightarrow \neg q$ ② $q \rightarrow r$ ③ $\neg r \rightarrow \neg p$ is True

By defn. of conditional, in (1) if p is True, $\neg q$ is True. By defn. of negation, q is False. By defn. of conditional, $q \rightarrow r$ is True which is consistent with (2). Since p is True, $\neg p$ is False by defn. of negation. By defn. of conditional, $\neg r$ is False (3). By defn. of negation, r is True.

One possibility: $\boxed{p \text{ True } q \text{ False } r \text{ True}}$

Let p is False. ① is consistent (by defn. of conditional) $\neg p$ is True (by defn. of negation). By ③, defn. of conditional, $\neg r$ is True. False. If $\neg r$ is True, r is False (by defn. of negation). By defn. of conditional (2), q is False. If $\neg r$ is False, r is True (by defn. of negation)

It is consistent with ②. $\therefore$ The specifications are consistent.

11. Are these system specifications consistent? "The router can send packets to the edge system only if it supports the new address space. For the router to support the new address space it is necessary that the latest software release be installed. The router can send packets to the edge system if the latest software release is installed. The router does not support the new address space."

Soln: p : The router can send packets to the edge system
 q : The router supports the new address space
 r : The latest software release be installed.

① $p \rightarrow q$ ② $q \rightarrow r$ ③ $r \rightarrow p$ ④ $\neg q$ are True

By defn. of negation, q is False. By defn. of conditional, $q \rightarrow r$ is True. (consistent with 2)